

Köhler illumination in the TEM: fundamentals and advantages

G. BENNER & W. PROBST

Electron Optics Division, Carl Zeiss, 73446 Oberkochen, Germany

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Summary

Köhler illumination is the most favourable design for the illumination path of an electron microscope with a condenser objective lens. The new illumination system of the EM 910 and EM 912 OMEGA allows both wide area (Köhler) illumination for TEM operation and spot illumination for analytical investigations. Compared to conventional systems and objective lenses with a condenser mini lens, this system offers many advantages. In addition to the homogeneous, highly coherent and parallel illumination of every point in the specimen, it offers advantages for selected area diffraction and spot scan mode.

Combined with the electron optical selection of a condenser aperture, this illumination system provides the flexibility necessary to achieve optimum illumination for the specimen.

Introduction

The most suitable objective lens for an analytical transmission electron microscope (TEM) is the single-field condenser objective lens, first developed by Riecke & Ruska (1966), in which the specimen is located at the centre of the polepiece gap. This approach permits both high-resolution TEM operation and the generation of small electron probes needed for the analytical, microdiffraction and scanning transmission electron microscope (STEM) modes of operation. The specimen position in the plane of symmetry of the condenser-objective lens also guarantees maximum specimen tilt capability and an optimum detector arrangement for energy-dispersive X-ray spectroscopy (EDX), secondary electron imaging (SE), and back-scattered electron imaging (BSE). In combination with the appropriate Köhler illumination system, this type of objective lens offers the most flexible capabilities for TEM and spot illumination.

An optimum illumination system for use in an analytical TEM should meet the following requirements:

1 variable illumination angle to allow spatial coherence, as

well as the image brightness and contrast to be adapted to any particular specimen;

2 an illuminated specimen area only slightly larger than the imaged area in order to prevent unnecessary radiation damage to the specimen and to avoid stray electrons which would reduce contrast;

3 perpendicular beam incidence at every point of the specimen to prevent image distortion;

4 uniform and homogeneous illumination of the field of view;

5 simple switchover from TEM to spot (STEM) operation; and

6 separate adjustment of the illumination tilt and shift functions.

Only Köhler illumination can fulfil all of these requirements. This paper describes the principles and advantages of this new illumination system.

Types of objective lenses and illumination systems

In the classical TEM design the specimen is located close to the upper pole piece of the objective lens; that means the magnetic field above the specimen is very low and cannot influence the illuminating electron beam. Using a fixed condenser aperture, brightness and illuminated area can easily be adjusted by the focus control of the final condenser lens. For high-brightness illumination the cross over (CO) of the electron gun is imaged on the specimen and the illuminating angle σ is defined by the size of the condenser aperture. To illuminate a large specimen area the CO is focused by the final condenser in the condenser aperture. In this case the illuminating angle is given by the diameter of the demagnified CO image, but the illuminated area is not limited by the condenser aperture. The ray paths for focused illumination (solid lines) and wide-area illumination (dashed lines) of the specimen are shown in Fig. 1(a).

The main advantages of the classical approach are:

1 easy control of the brightness by the final condenser lens;

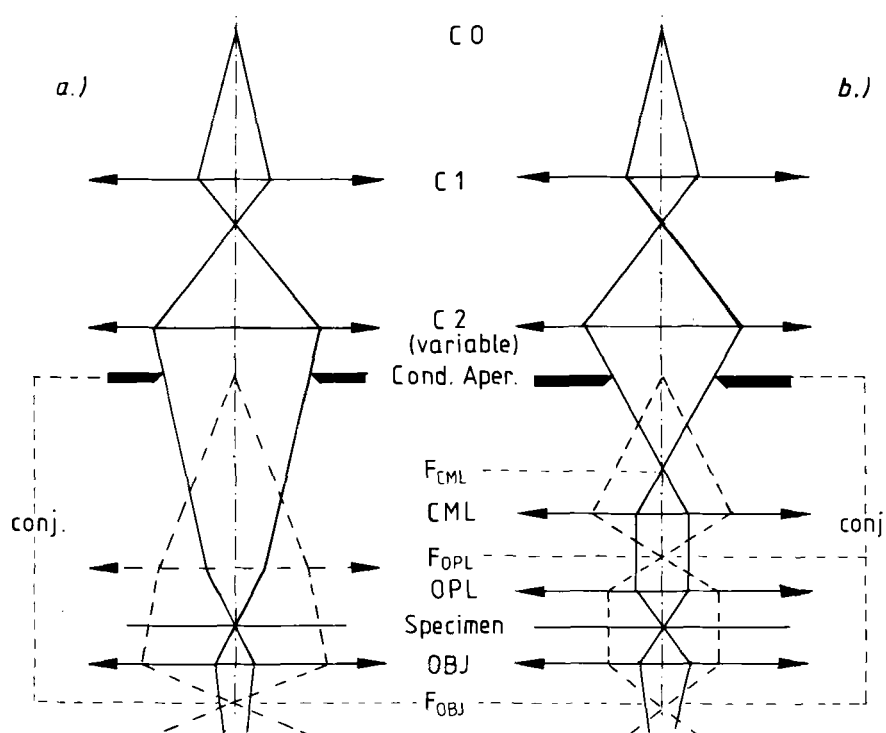


Fig. 1. Ray path of a classical illumination system (a) and a system with a condenser mini lens (CML) (b). In both systems the condenser aperture is conjugated to the back focal plane F_{OBJ} of the objective lens. For focused illumination the cross-over (CO) will be directly imaged on the specimen or with a CML via an intermediate CO image in the front focal plane (F_{CML}) of the auxiliary lens (solid lines). To illuminate a large area of the specimen the CO image has to be focused by the final condenser into the condenser aperture plane (dashed lines).

- 2 no condenser aperture exchange is required;
- 3 simple design of the illumination system—only two condenser lenses are necessary; and
- 4 very stable alignment—only minimum adjustment by the user is required.

But there are also severe restrictions:

- 1 no independent selection of illuminating angle (brightness) and illuminated field;
- 2 limited eucentric tilt capability of the specimen because of the interference with the upper pole piece; and
- 3 no high-current small spots can be generated, because of the absence of a strong focusing condenser field above the specimen.

To overcome these restrictions the specimen has to be placed in the centre of the pole piece gap of the objective lens. The so-called single field condenser-objective lens designed by Riecke & Ruska (1966) offers two advantages in comparison with the traditional objective lens. Due to the strong magnetic field at the specimen (a) the electrons are kept close to the optical axis resulting in lower lens aberration coefficients, and (b) the upper part of the objective lens field forms a strong condenser lens able to focus a small spot.

Another problem arises with the condenser-objective lens design, illustrated in Fig. 2. If the CO image is focused into the front focal plane of the objective prefield lens (OPL) (solid lines) the whole specimen is illuminated parallel to the optical axis and the illuminated field is defined by the condenser aperture. If, however, the final condenser is excited more to obtain a higher intensity, the CO image

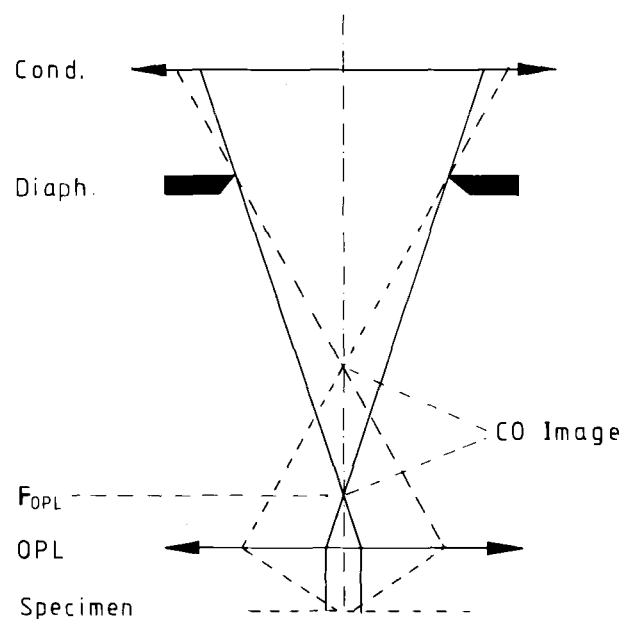
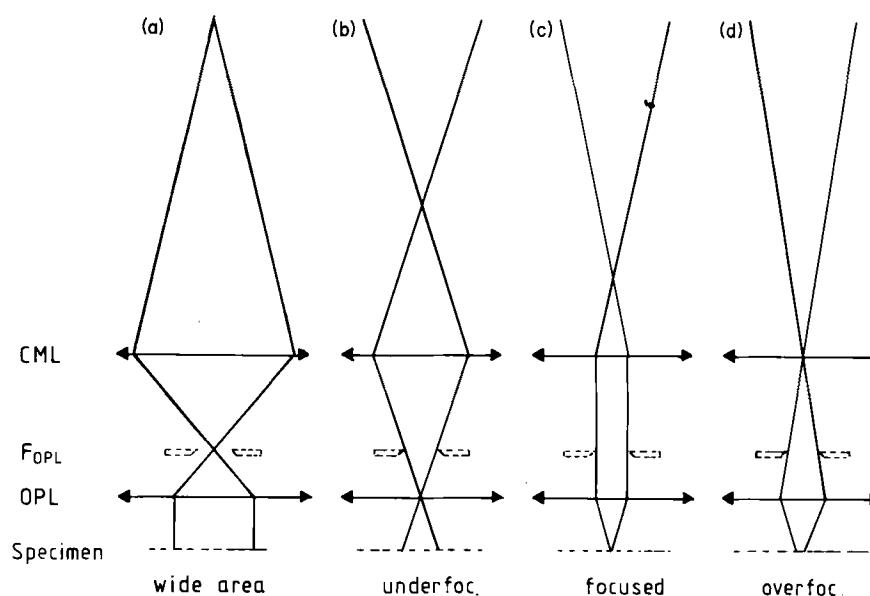


Fig. 2. The effect of final condenser lens variations for a condenser objective lens (COL). Only if the cross-over (CO) image is focused into the front focal plane (F_{OPL}) of the objective prefield lens (OPL) (solid lines) is the whole specimen illuminated parallel to the optical axis. Due to the strong focusing effect of the prefield lens, the off-axis specimen areas will be illuminated by strongly tilted electron beams if the condenser lens focus is changed (dashed lines).

Fig. 3. Tilted illumination due to final condenser variation in a system with condenser mini lens (CML). If the cross-over is not imaged in the front focal plane (F_{OPL}) of the objective prefield lens (OPL), or if the cross-over is not exactly focused on the specimen (a and c), then the sample will be illuminated by a divergent (b) or convergent (d) beam.



moves closer to the condenser system. Due to the strong focusing effect of the prefield lens, off-axis specimen areas will then be illuminated by strongly tilted electron beams. This means, that it is no longer possible to adjust the brightness by varying the focal length of the final condenser lens. The condenser aperture is conjugate with the specimen plane and acts as a field-limiting aperture for the illuminated area.

One solution to overcome this effect is to introduce an auxilliary condenser mini lens (CML) above the condenser objective lens (Van der Mast *et al.*, 1980) which images the condenser diaphragm into the front focal plane of the prefield lens, and the condenser diaphragm has the same effect as in a classical illumination system.

Figure 1 shows the ray path of a classical illuminating system (Fig. 1a) and a system with a CML (Fig. 1b). In both systems the condenser aperture is conjugate with the back focal plane of the objective lens (i.e. the image of the condenser aperture can be seen when the instrument is switched to diffraction mode). The size of the beam diameter in the condenser aperture plane defines the illuminating angle.

For focused illumination the CO will be directly imaged on the specimen or with a CML via an intermediate CO image in the front focal plane of the auxilliary lens (Fig. 1, solid lines). To illuminate a large area of the specimen the CO image has to be focused by the final condenser into the condenser aperture plane (Fig. 1, dashed lines).

Although this concept avoids some of the limitations of a conventional objective lens it has still a few disadvantages.

- 1 There is no independent adjustment of the illumination angle and the illuminated field.
- 2 The illumination is not homogeneous, particularly if the CO is imaged on the specimen.
- 3 The off-axis incident electron beam is not parallel to the

optical axis in the general case. This can be seen in Fig. 3. If the CO is not imaged in the front focal plane of the objective prefield lens or if the CO is not exactly focused on the specimen then the sample will be illuminated by a divergent or convergent beam.

4 In order to obtain spot diameters in the nanometre range the CML has to be switched off, requiring a readjustment of the illumination system.

5 The scanning coils cannot be placed close to the objective prefield lens because of the space requirements of the CML.

A better solution for driving a condenser-objective lens is the introduction of Köhler illumination.

Principles of Köhler illumination

The requirement and properties of a condenser-objective lens are ideally complemented by the Köhler illumination system. This illumination concept was proposed by August Köhler (1893) one hundred years ago for the light microscope (LM).

The two essential points of a Köhler illuminating system are:

- 1 imaging of the (electron) source into the front focal plane of the objective prefield lens (OPL) by the condenser system; and
- 2 imaging of the condenser aperture into the specimen plane by the objective prefield lens.

Figure 4 shows the illumination cones for the axial point and the right and left edge of the illuminated area. Due to the position of the CO image in the front focal plane, the incident electron beam is perpendicular at every point of the specimen (parallel illumination). The illumination angle σ is determined by the radius of the source image r_{CO} divided by the focal length f_{OPL} of the OPL.

$$\sigma = r_{CO}/f_{OPL}$$

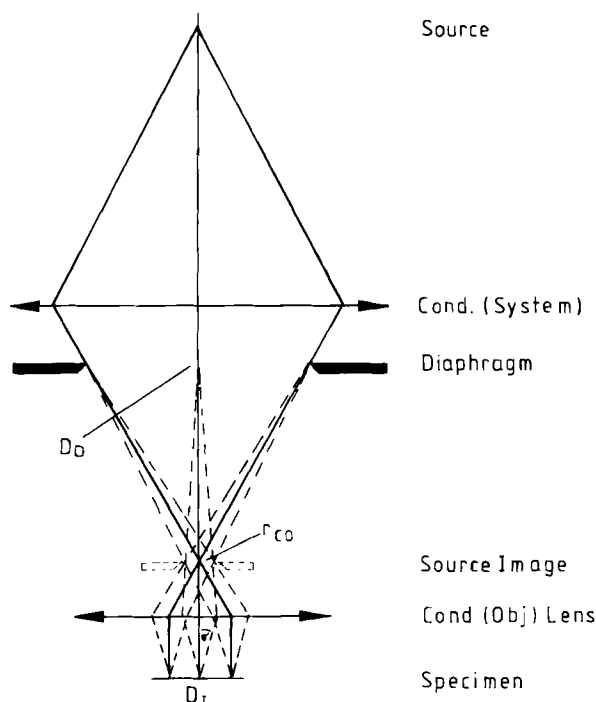


Fig. 4. The principle ray path of Köhler illumination. Due to the position of the source image in the front focal plane of the condenser (objective) lens the incident electron beam is perpendicular at every point of the specimen (i.e. parallel illumination). The illumination angle σ is determined by the radius of the source image r_{CO} and the illuminated area by the diameter of the condenser diaphragm.

The size of the illuminated area D_I depends only on the diameter of the condenser diaphragm D_D and the (de)magnification M_{OPL} provided by the OPL.

$$D_I = D_D M_{OPL} \quad \text{with} \quad M_{OPL} < 1.$$

The size of the condenser aperture does not affect the size of the source image and vice versa; this means that the illuminated area and the illuminating angle can be selected independently of each other.

In a light microscope, the size of the condenser aperture and the size of the source image can be easily changed by using two separate, adjustable iris-type diaphragms. However, in an electron microscope there are no variable diaphragms available and the realization of a small illuminating angle (≤ 0.2 mrad) requires a CO diameter of less than $1 \mu\text{m}$.

This 'aperture problem' demands a special technical solution for the implementation of Köhler illumination in a TEM, but does offer additional opportunities.

Technical realization of Köhler illumination in a TEM

The illumination systems of Benner *et al.* (1990) consist of four magnetic lenses—three condenser lenses and the objective prefield lens. Figure 5 shows the beam path for TEM and spot (STEM) operation. The first and second condensers are operated as zoom systems; this means that they image the CO of the gun at variable (de)magnifications into the spatially fixed zoom image plane which is congruent to the plane containing the pivot point of the condenser deflection system. In the TEM mode, the 3rd condenser transfers this CO image into the front focal plane of the OPL. Above the OPL is situated a double deflection system, the pivot point of which lies in the front focal plane of the OPL thus enabling the illumination to be shifted without the need to tilt the incident electron beam. The central plane of the upper deflection coils of the double deflection system is conjugate with the specimen plane. A deflection of the beam by this system therefore causes only a beam tilt. The condenser diaphragm is located in the central plane of the third condenser lens; in TEM mode this diaphragm acts as a field-limiting aperture. In the spot (STEM) mode, the 3rd condenser lens images the CO into the entrance image plane of the OPL, generating a demagnified image of the CO on the specimen. In this case, the condenser diaphragm limits the aperture of the focused spot. The illumination angle and the spot size can be easily changed by varying the demagnification of the condenser zoom system.

The size of the illuminated area should be matched to the magnification of the EM for optimum TEM illumination. This can be achieved automatically by using the automatic illumination selector (AIS). Figure 6 shows the setup of the AIS system.

Depending on the magnification used in the EM, the electron beam is deflected by the condenser deflection system to one of the openings in a multi-hole condenser diaphragm. The beam is focused by the third condenser on to the front focal plane of the OPL and shifted back to the optical axis by the illumination deflection system. At high TEM magnifications and in the spot mode, the AIS is switched off, resulting in the use of the central aperture. In this way, AIS allows the simple changeover from TEM- to spot-operation. Independently of the selected magnification any condenser aperture can be manually selected in 'manual illumination select' (MIS) mode, e.g. for the purpose of selected area (SA) diffraction or for the selection of a small area to focus the image.

Practical advantages for TEM operation

Such a carefully designed illumination system provides even more advantages in the TEM than in the LM (Probst *et al.*, 1991).

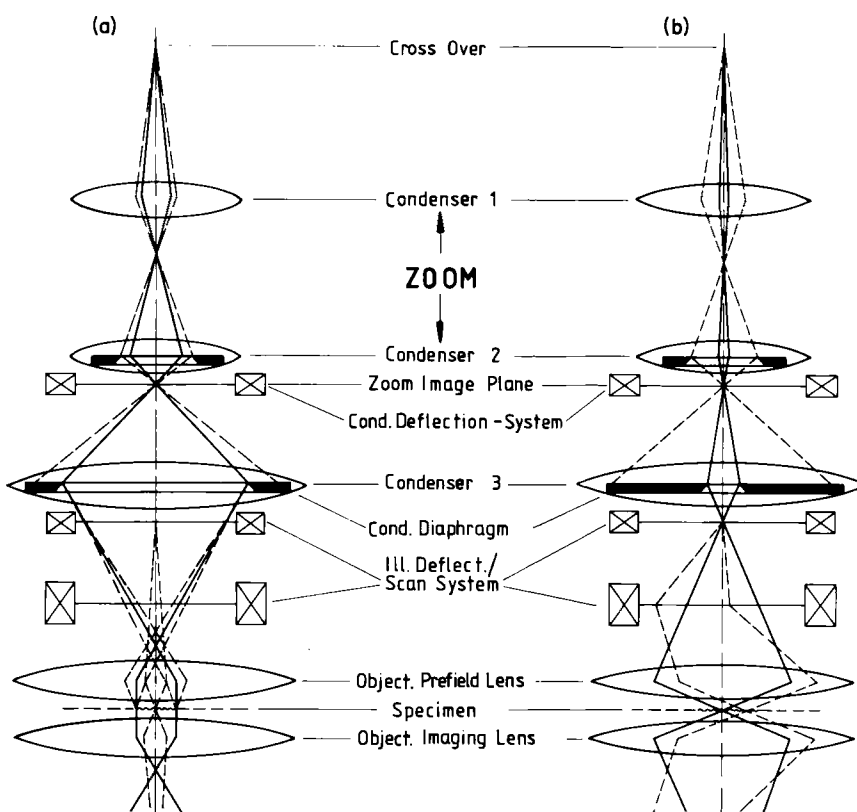


Fig. 5. Ray paths for Köhler (a) and spot (b) illumination in a TEM. The 1st and 2nd condensers image the cross-over of the gun with variable magnification into the fixed zoom image plane. In TEM mode, the 3rd condenser transfers this CO image into the front focal plane of the objective prefield lens (OPL) and the condenser diaphragm acts as a field-limiting aperture. In the spot (STEM) mode, the 3rd condenser images the CO into the entrance image plane of the OPL, generating a demagnified image of the CO on the specimen. In this case, the condenser diaphragm limits the aperture of the focused spot.

Firstly, the effects of illuminating aperture on spatial resolution and depth of field can easily be varied and selected independently of the illuminated area. Due to the damping of the contrast transfer function (CTF) with increasing illumination angle it is very important

to select a small illuminating angle in order to achieve the optimum resolution. To demonstrate this effect a carbon foil was illuminated with an astigmatic CO image. The illuminating angle in one direction was 0.25 mrad and in the other direction about three times larger. Figure 7

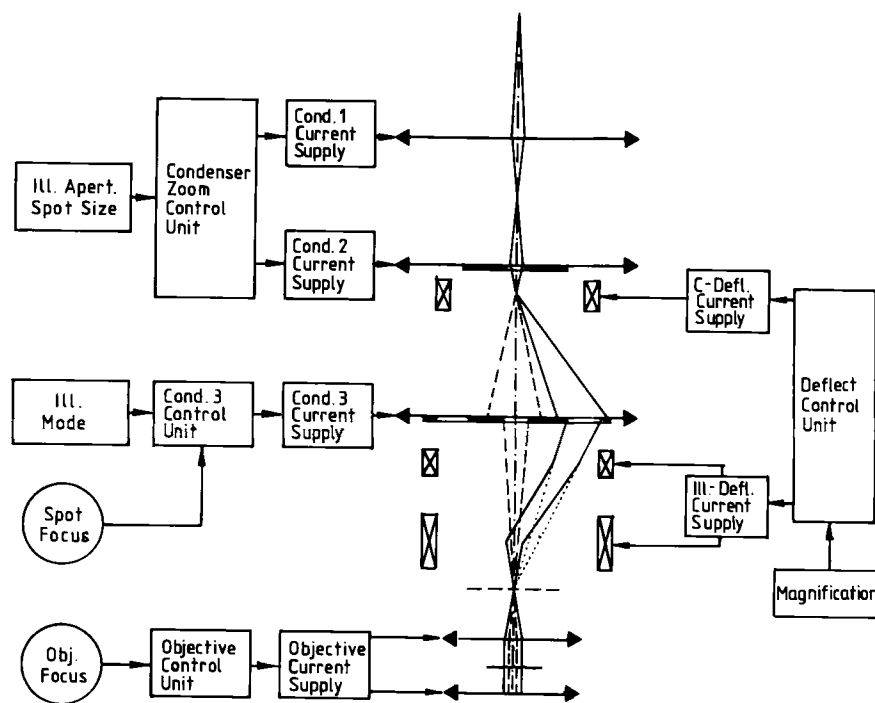


Fig. 6. Setup of the automatic illumination selector (AIS). Depending on the magnification used in the EM, the electron beam is deflected by a condenser deflection system to one of the openings in a multi-hole condenser diaphragm and shifted back to the optical axis by the illumination deflection system.

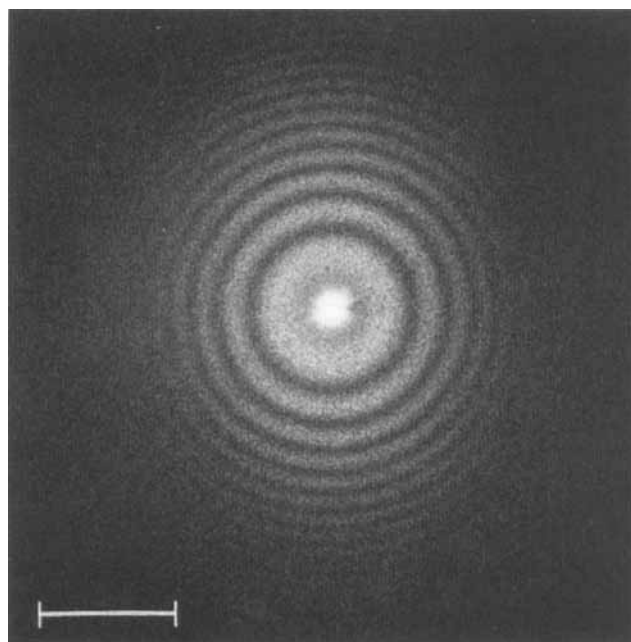


Fig. 7. Diffractogram of a carbon foil illuminated with an astigmatic cross-over. The illumination angle in one direction is 0.25 mrad and in the other direction about three times larger. The magnification is $50\,000\times$ and the defocus is $1 \mu\text{m}$. In one direction up to 10 fringes can be counted but only six fringes in the other direction. Scale bar = 1 nm .

shows the optical diffractogram of the micrograph taken at a magnification of $50\,000:1$ and a defocus of $1 \mu\text{m}$. In the smaller-angle direction it is possible to count up to 10 fringes but in the other direction only six fringes can be seen.

Secondly, because the ray pencils are parallel to the optical axis, distortion-free phase contrast images and diffraction patterns are obtained over the whole micrograph. To demonstrate this, diffractograms of a high resolution image at low magnification ($50\,000:1$) were taken at various locations (circles in Fig. 8). The diffractograms at the corners show the same shape as the central diffractogram (Fig. 9), which means there are no changes in focus and beam tilt.

Thirdly, the illumination of every specimen point by a pencil of rays coming from the CO image in the focal plane of the OPL results in homogeneously illuminated crisp photographs, independent of the filament saturation. As an example of this excellent performance, Fig. 10 shows a micrograph of a spread collagen molecule.

Finally, the illuminated field is automatically adapted to the size of the viewing field, thus optimizing image contrast due to reduction of stray electrons. Moreover, irradiation damage of the specimen is minimized and restricted to the imaged area independent of any variation in brightness.

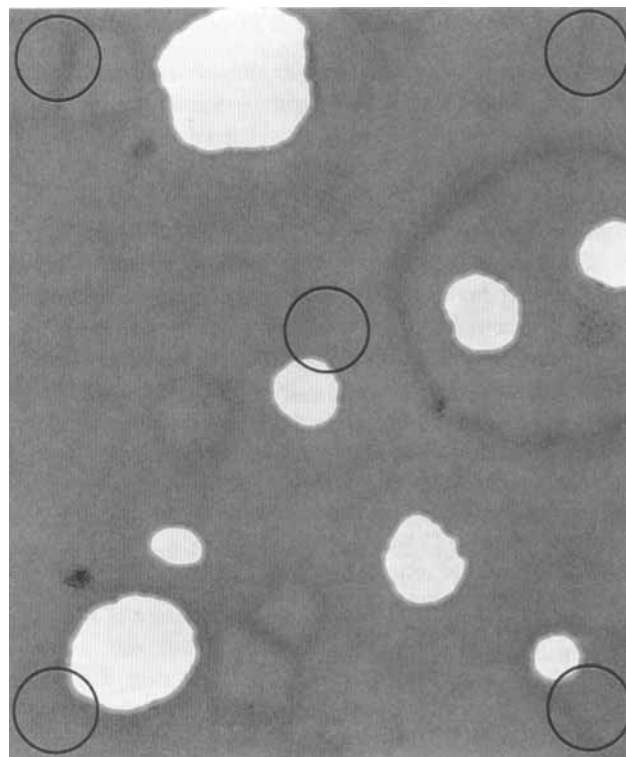


Fig. 8. High-resolution image of a carbon foil taken at a magnification of $50\,000\times$ and a defocus of $1 \mu\text{m}$ with an illumination angle of 0.25 mrad . The circles mark the positions where the diffractograms were taken (see Fig. 9).

Advantages for selected-area diffraction

Köhler illumination also provides advantages for selected-area (SA) diffraction. To obtain the coherent illumination required for SA diffraction in a TEM without Köhler illumination the CO has to be focused into the condenser aperture plane, resulting in the illumination of a large area of the specimen. The diffracting area therefore must be limited by an SA aperture in the intermediate image plane (Fig. 11a).

Because of the spherical aberration of the objective lens, the higher order reflections of the SA may be intercepted by the SA aperture, and may thus be lost in the photograph of the diffraction pattern. However, high-order rays from adjacent specimen areas may pass the SA aperture. This limits the minimum selectable area in the conventional SA diffraction mode to approximately $1 \mu\text{m}$ (Agar, 1960). With Köhler illumination (Fig. 11b), the illuminated area is determined by the condenser aperture, i.e. the illuminated and the diffracting area are identical. Thus the size of the SA is limited only by the diameter of the smallest available condenser aperture. For example, a condenser aperture of $5 \mu\text{m}$ and a demagnification of $30:1$ for the objective prefield lens results in a minimum

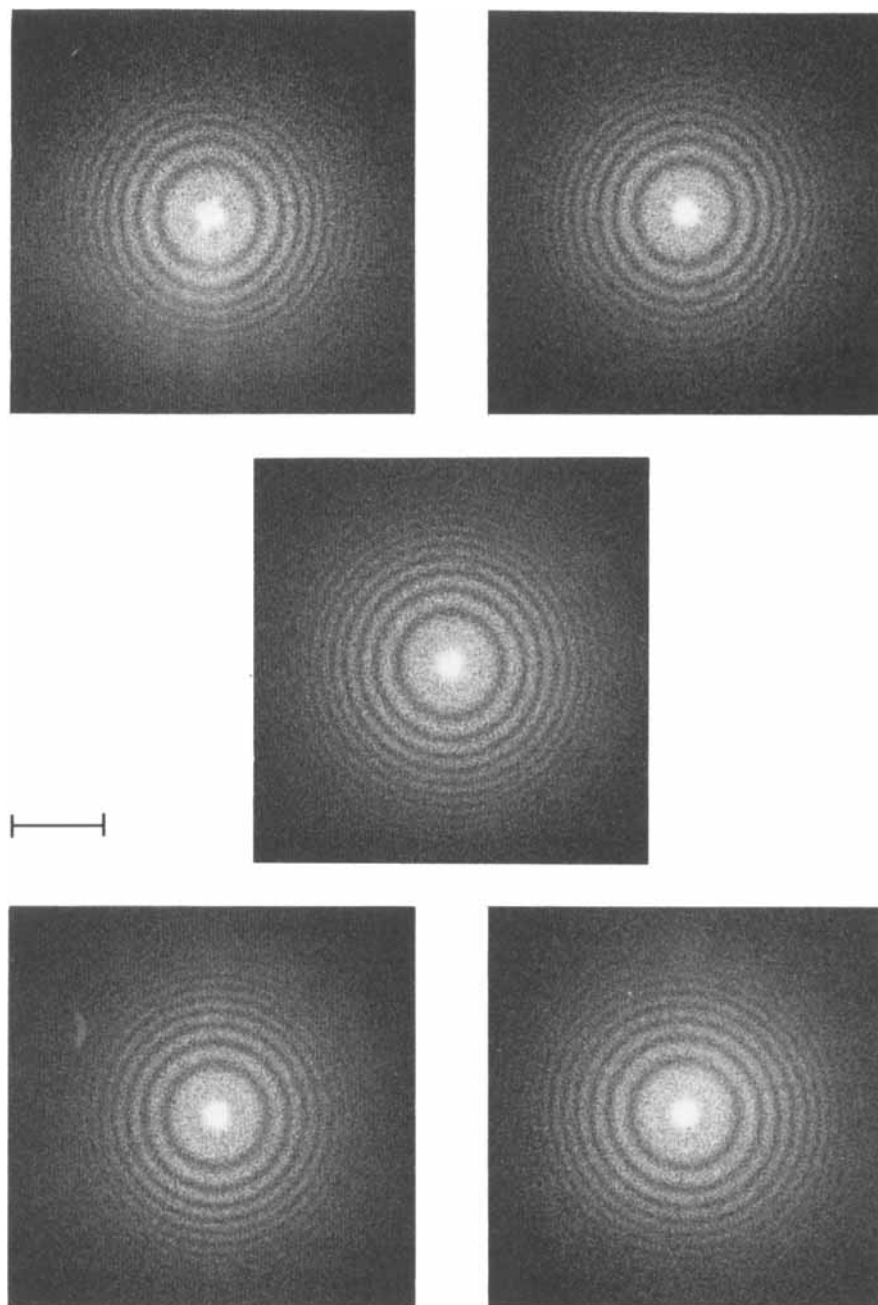


Fig. 9. Diffractograms near the centre and the corners of the carbon foil marked in Fig. 8. The diffractograms at the corners show the same shape as the central one, indicating that there are no changes in the focus and beam tilt. Scale bar = 1 nm.

diffraction area of 180 nm in diameter. The size of the diffraction spots can be set by the demagnification of the CO image using a suitable excitation of the condenser zoom system, independently of the selected area.

Figure 12 shows an example for chrysotile. If the SA of $0.7\ \mu\text{m}$ is positioned as shown in Fig. 12(a), a diffraction pattern of the two crystals is obtained (Fig. 12b). Figure 12(c,d) shows the diffraction pattern if the area is shifted only onto the right crystal. When selecting the diffracting area with an SA aperture, additional diffraction spots (marked by an arrow) from the neighbouring crystal (Fig.

12c) appear in the diffraction pattern. These diffraction spots are missing in the diffraction pattern obtained with Köhler illumination (Fig. 12d).

Using a $5\text{-}\mu\text{m}$ condenser aperture also provides the Micro Diffraction Mode. In the Spot illumination mode diffraction areas down to $1.6\ \text{nm}$ can be selected.

The Köhler illumination system also allows us to obtain diffraction patterns from beam sensitive specimens (polymers), because there is no need for large area illumination when performing selected area diffraction.

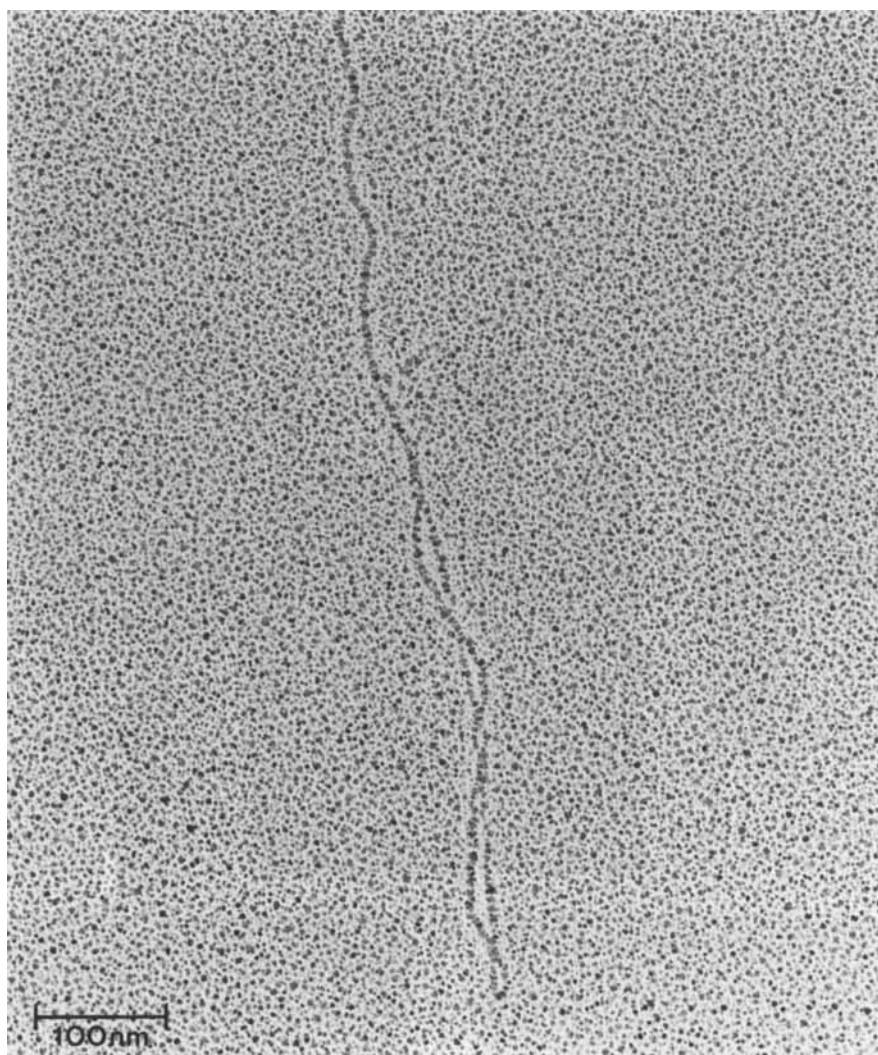


Fig. 10. An extended collagen molecule. The illumination of every specimen point by a pencil of rays coming from the cross-over point results in a homogeneously illuminated, crisp micrograph.

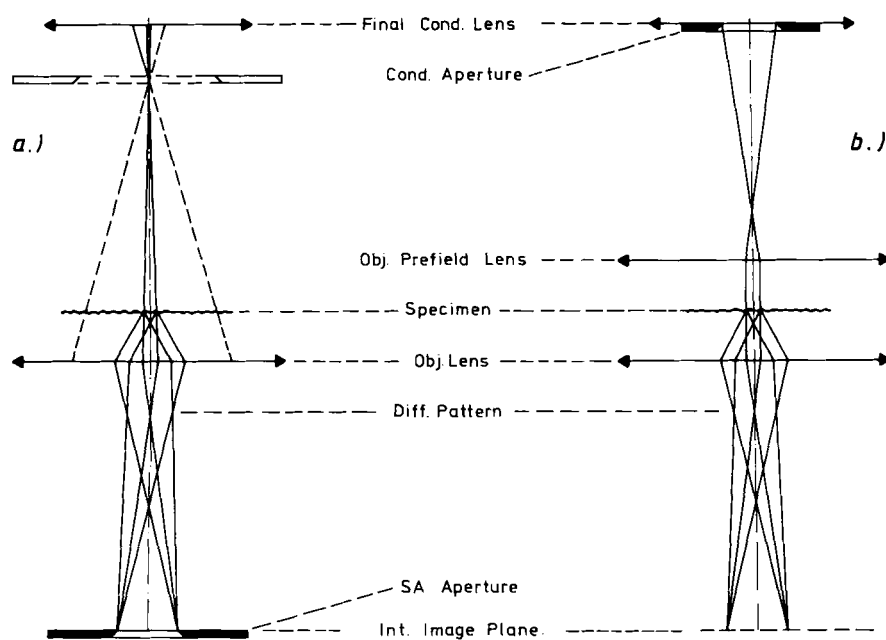


Fig. 11. Ray path for selected-area (SA) diffraction for conventional (a) and Köhler (b) illumination. To obtain coherent illumination in a TEM without Köhler illumination the cross-over has to be focused into the condenser aperture plane, resulting in the illumination of a large area of the specimen. The diffracting area therefore must be limited by an SA aperture in the intermediate image plane. With Köhler illumination, the illuminated area is determined by the condenser aperture, i.e. the illuminated area and the diffracting area are identical.

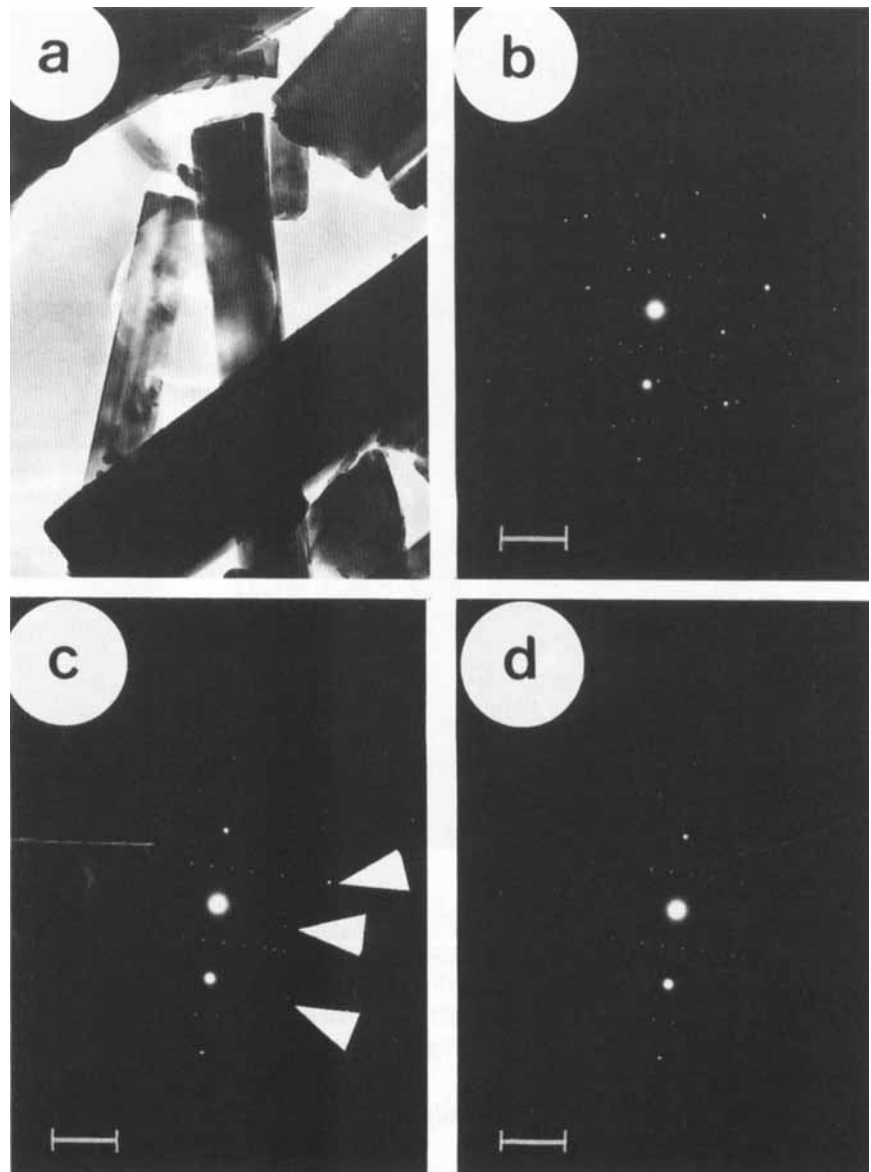


Fig. 12. Example demonstrating the advantages of Köhler illumination for selected-area (SA) diffraction. If the SA of $0.7\ \mu\text{m}$ is positioned as shown in Fig. 2(a) a diffraction pattern of the two crystals is obtained (b). If the SA is shifted only on the right crystal a diffraction pattern, as in (c) or (d), appears. If the diffracting area is selected by an SA aperture, additional diffraction spots (arrows) coming from the neighbouring crystal (c) appear in the diffraction pattern. These diffraction spots are missing in the diffraction pattern obtained with Köhler illumination (d). Scale bars = $5\ \text{nm}$.

Advantages for spot and slit scan illumination

In the spot scan mode described by Bullough & Henderson (1987), the photographic plate is exposed in a step-scan process by using a small illuminating spot. Spot scanning in TEM reduces specimen damage and specimen drift. In addition to this, it allows dynamic focusing at high tilt angles of the specimen.

High-resolution, 3-D specimen reconstruction (de Jong & Typke, 1990) and reflection electron imaging using specimen tilt angles of up to 90° require dynamic focusing perpendicular to the tilt axis of the goniometer. A slit-type illumination of the specimen parallel to the goniometer tilt axis may provide a better solution for such problems.

In the Köhler illumination system the condenser aperture acts as an illuminated field-limiting aperture. In addition to sharp illumination spots, Köhler illumination allows slit illumination. With a slit aperture of $\leq 3\ \mu\text{m}$ width, illuminated fields on the specimen with a width of less than $100\ \text{nm}$ can be generated. Instead of condenser apertures with fixed slit width, a slit aperture with a variable slit width can also be used. Hence, the slit width can be optimally adapted to the microscope magnification and to the tilt angle used.

Figure 13 shows a micrograph of ferritin illuminated with hexagonally scanned $200\ \text{nm}$ spots. Figure 14 shows an example for the slit scan mode. The picture of a 75° tilted specimen was taken without focus correction. The slit width of the overlapping slits images is $100\ \text{nm}$. The clearly visible

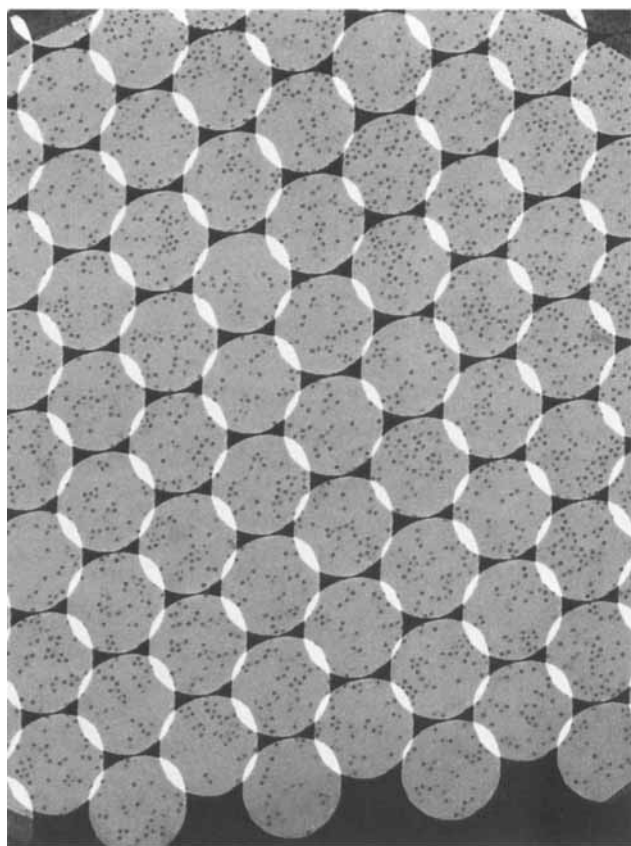


Fig. 13. Hexagonal spot scan illumination. In a Köhler illumination system the condenser aperture acts as an illumination field-limiting aperture, resulting in sharp illumination spots. Spot diameter = 200 nm.

focus change perpendicular to the slits indicates the necessity for dynamic focusing.

The slit scan mode offers the following advantages over the spot scan mode.

- 1 Complete exposure of the photographic plate without any loss of information.
- 2 Because of the exposure of slit-shaped specimen areas at a defined tilt angle and focus level, the exposure times are shorter and specimen drift is reduced.

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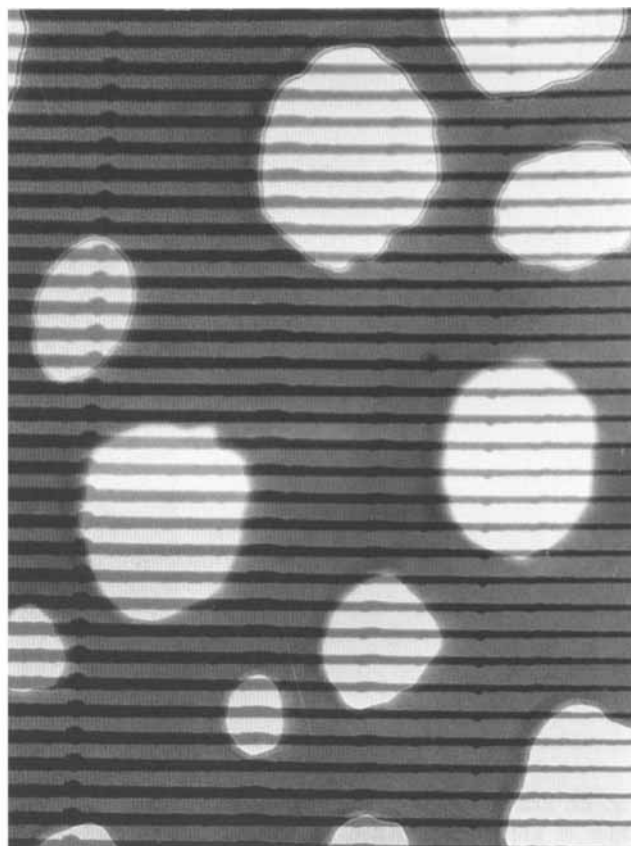


Fig. 14. Overlapping slit scan illumination of a specimen tilted at 75° without focus correction. The clearly visible focus change perpendicular to the slits illustrates the necessity for dynamic focusing. The slit widths of the overlapping slit images are 100 nm.